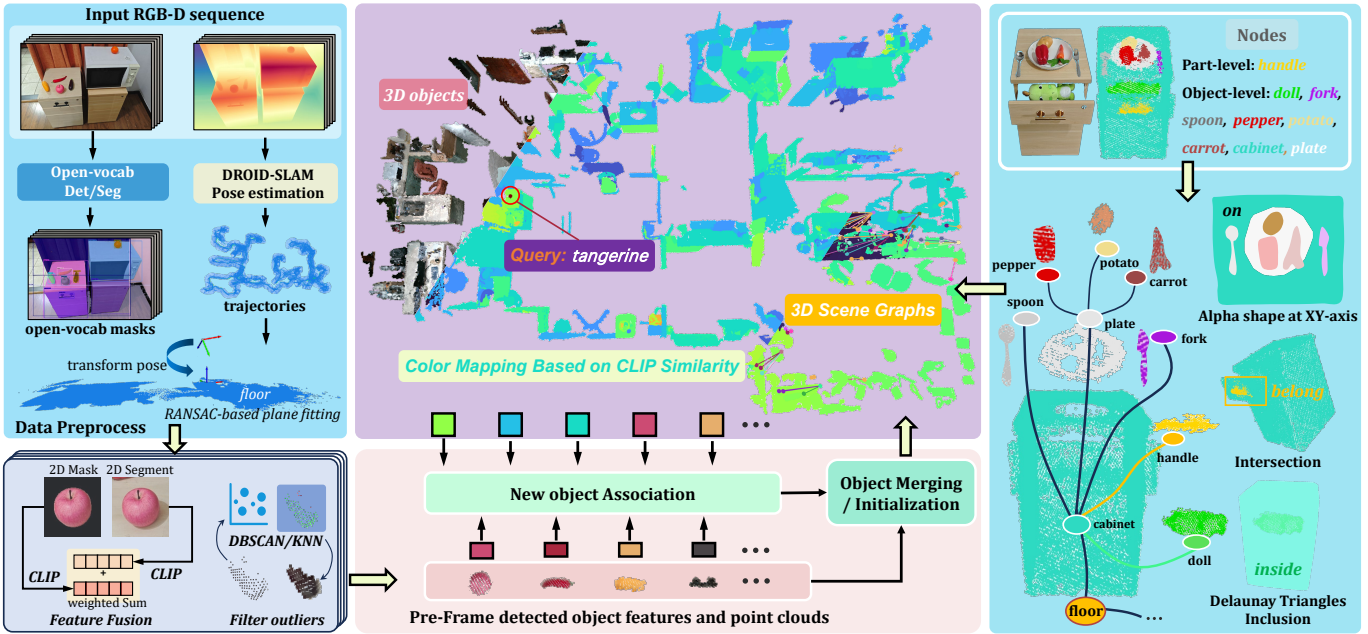

Input RGB-D sequence
Open-vocab Det/Seg
DROID-SLAM Pose estimation
open-vocab masks
trajectories
transform pose
floor
RANSAC-based plane fitting
Data Preprocess
2D Mask
2D Segment
CLIP
CLIP
weighted Sum
Feature Fusion
DBSCAN/KNN
Filter outliers
3D objects
Query: tangerine
3D Scene Graphs
Color Mapping Based on CLIP Similarity
New object Association
Object Merging / Initialization
Pre-Frame detected object features and point clouds
Nodes
Part-level: handle
Object-level: doll, fork, spoon, pepper, potato, carrot, cabinet, plate
pepper
potato
carrot
spoon
plate
fork
handle
cabinet
doll
floor
on
Alpha shape at XY-axis
belong
Intersection
inside
Delaunay Triangles Inclusion